Bilkent University
Computer Engineering Department
CS 491 Senior Design Project I

Assessment of Innovation Form

ID and Title of the Project: SENTINEL: Surveillance, Explainability, and Novelty Tracking in INvestment and Equity Limits

Team Member IDs: 22301879 - 22102837 - 22203553 - 22201729

Name of the Supervisor: Selim Aksoy

Below questions needs to be filled by the Innovation Expert

*Considering the definition of innovation as "**creation of better or more effective products, systems, services, or technologies that have the potential to be accepted by markets, governments, and society**", please assess the innovativeness of the design project with below questions.*

1) The project and subject that is proposed and presented to you is:

Criteria	Poor	Unsatisfactory	Satisfactory	Good	Outstanding
Technically feasible:	☐	☐	☐	☐	☑
You see enough market demand:	☐	☐	☐	☐	☑
Development planning is done well	☐	☐	☐	☑	☐
Enough research done for marketplace and competitors	☐	☐	☐	☑	☐
Delivers enough value or solves a real problem	☐	☐	☐	☑	☐

2) What is the nature of the innovation you see in this project? Do you have any suggestion to improve converting the knowledge and idea of the subject more into benefit and value?

The most prominent innovation I see in the project is that it offers a context-aware framework that detects deviations from the historical and benchmark behaviors of funds, rather than relying on an existing infrastructure based on fixed rules and thresholds.

The fact that it includes a complete service-layered project architecture, including a user interface, is very valuable.

Continuously improving accuracy by feeding expert review results back into the algorithm could add significantly more value to the project and enhance its sustainability.

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3) Are there any unaddressed risks that team members need to consider during implementation?

Incompatible data formats coming from different sources may lead to delays or excessive effort during development processes.

There is a possibility that the algorithm might overfit historical data and generate false positive alarms.

There is a risk that the algorithm may prove inadequate or produce erroneous results for newly onboarded funds or those with limited historical data.

4) Any suggestions while shaping go to market strategy?

Real-time comparison (A/B testing) on live data should be performed by running the existing rule-based FGP in parallel with the new algorithm.
It should also feature an interface where experts can log their decisions (which will assist in refining the algorithm in the future).
Anomaly detection services and risk scoring services should be designed as independent microservices; this prevents scaling issues and allows for differentiated pricing per product (if applicable).
It must be ensured that data is anonymized and security is maintained at the highest level.

Innovation Expert

Name: …Abdullah Sönmez……

Date:…22……/…10…/……2026……

Signature:…Asonmez……

Final Expert Score:….4….. (out of 5)

Scale:
(1) Poor, (2) Unsatisfactory, (3) Satisfactory, (4) Good, (5) Outstanding